

No. of Functional Features Included in the Solution

Date	20 July 2026
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine

Functional Features Overview

S.No	Feature Name	Feature Description	Module/Component	Status	Marks Contribution
		Loads			
1	Dataset Loader & Cleaning	<code>crop_recommendation.csv</code> , removes duplicates and missing values	<code>train_model.py</code>	Done	—
2	Outlier Removal (IQR)	Detects and removes outliers per feature using the Interquartile Range method	<code>train_model.py</code>	Done	—
3	Exploratory Data Analysis	Boxplots (univariate), correlation heatmap (bivariate), pairplot (multivariate)	<code>train_model.py</code> → <code>static/images/</code>	Done	—
4	Seasonal Crop Grouping	Rule-based grouping of crops into summer/winter/rainy categories for insight	<code>train_model.py</code>	Done	—
5	K-Means Clustering + Elbow Method	Unsupervised clustering of soil/climate profiles with optimal-k visualization	<code>train_model.py</code> → <code>static/images/elbow.png</code>	Done	—
6	Model Comparison (Logistic Regression vs. Random Forest)	Trains both, compares accuracy/precision/recall/F1, selects the best	<code>train_model.py</code>	Done	—
7	Model Persistence	Saves the best model + feature order to <code>model.pkl</code>	<code>train_model.py</code>	Done	—
8	Home Page	Landing page introducing OptiCrop	<code>templates/home.html</code>	Done	—
9	About Page	Objectives, methodology, and input feature explanation	<code>templates/about.html</code>	Done	—
10	Find Your Crop (Prediction) Page	Input form + real-time crop recommendation	<code>templates/findyourcrop.html</code> , <code>app.py</code>	Done	—

Feature Summary

Metric	Count / Value
Total Features Planned	10
Total Features Implemented	10
Core / Must-Have Features	7 (Data loading/cleaning, outlier removal, model comparison, persistence, prediction page)
Additional / Nice-to-Have Features	3 (Seasonal grouping, K-Means + Elbow, About page)
Features Tested & Verified	10

Feature Category Breakdown

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Home, About, Find Your Crop pages

2	Backend / Logic	2	Flask routing, input validation in <code>/predict</code>
3	Database / Storage	2	CSV dataset, serialized <code>model.pkl</code>
4	ML / Prediction	5	IQR outlier removal, EDA, K-Means + Elbow, Logistic Regression, Random Forest
5	Security / Authentication	0	Not applicable — single-session, stateless app